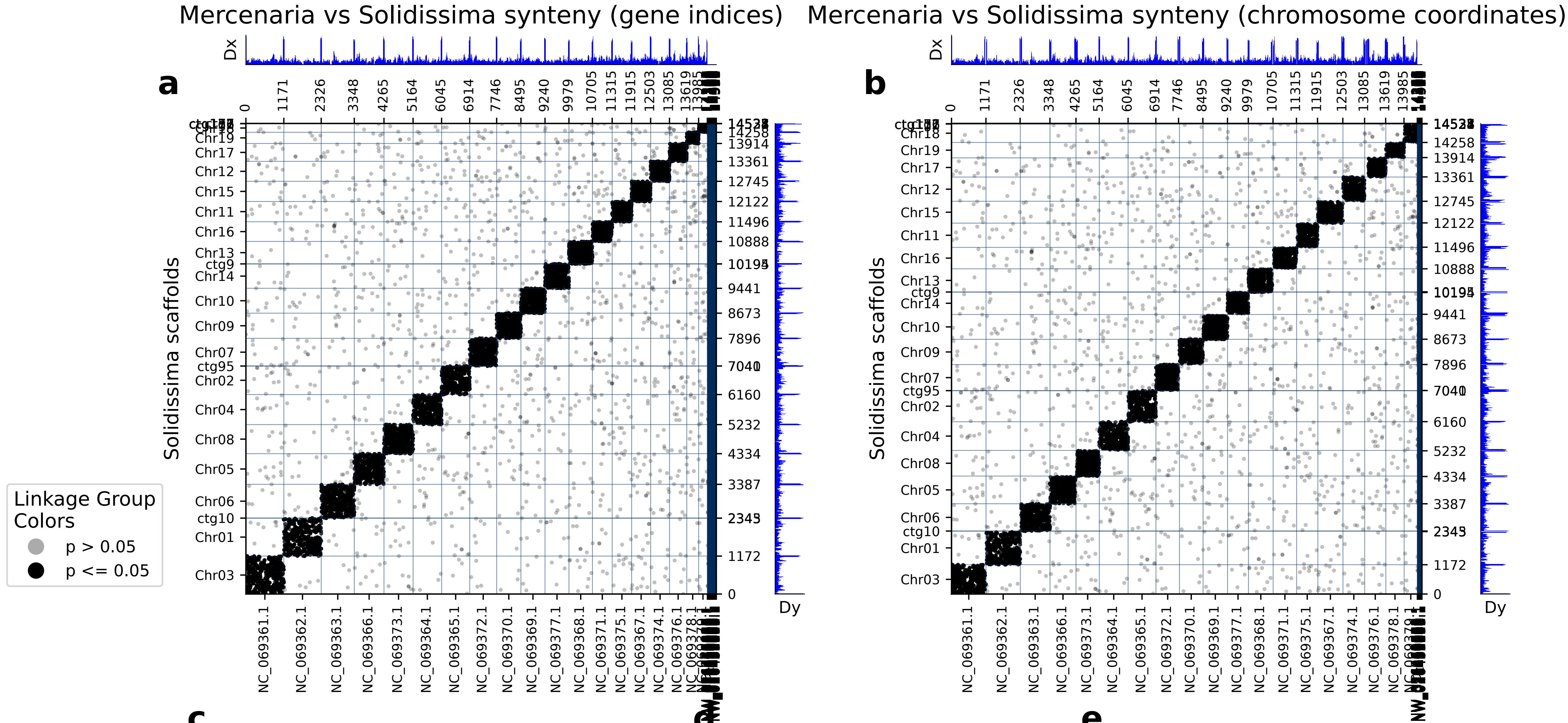
diamond: 14533 orthologs on 271 Mercenaria scaffolds and 27 Solidissima scaffolds.



c Mercenaria vs Solidissima groupings

d Mercenaria vs Solidissima FET, whole chroms.

	Mercenaria scaf	Solidissima scaf	p (FET)	Count
0	NC_069362.1	Chr01	0.00e+00	1071
1	NC_069361.1	Chr03	0.00e+00	1059
2	NC_069363.1	Chr06	0.00e+00	941
3	NC_069366.1	Chr05	0.00e+00	822
4	NC_069373.1	Chr08	0.00e+00	808
5	NC_069364.1	Chr04	0.00e+00	795
6	NC_069365.1	Chr02	0.00e+00	766
7	NC_069372.1	Chr07	0.00e+00	746
8	NC_069377.1	Chr14	0.00e+00	672
9	NC_069370.1	Chr09	0.00e+00	672
10	NC_069369.1	Chr10	0.00e+00	665
11	NC_069368.1	Chr13	0.00e+00	644
12	NC_069375.1	Chr11	0.00e+00	530
13	NC_069371.1	Chr16	0.00e+00	521
14	NC_069374.1	Chr12	0.00e+00	517
15	NC_069367.1	Chr15	0.00e+00	506
16	NC_069376.1	Chr17	0.00e+00	443
17	NC_069378.1	Chr19	0.00e+00	300
18	NC_069379.1	Chr18	0.00e+00	210
19	NC_048487.1	ctg177	5.01e-07	3
20	NC_048487.1	ctg116	1.46e-03	2

e Mercenaria vs Solidissima FET, chrom. pieces

	Mercenaria scaf	Solidissima scaf	p (FET)	Count
0	NC_069362.1 : 0-end	Chr01 : 0-end	0.00e+00	1071
1	NC_069361.1 : 0-end	Chr03 : 0-end	0.00e+00	1059
2	NC_069363.1 : 0-end	Chr06 : 0-end	0.00e+00	941
3	NC_069366.1 : 0-end	Chr05 : 0-end	0.00e+00	822
4	NC_069373.1 : 0-end	Chr08 : 0-end	0.00e+00	808
5	NC_069364.1 : 0-end	Chr04 : 0-end	0.00e+00	795
6	NC_069365.1 : 0-end	Chr02 : 0-end	0.00e+00	766
7	NC_069372.1 : 0-end	Chr07 : 0-end	0.00e+00	746
8	NC_069377.1 : 0-end	Chr14 : 0-end	0.00e+00	672
9	NC_069370.1 : 0-end	Chr09 : 0-end	0.00e+00	672
10	NC_069369.1 : 0-end	Chr10 : 0-end	0.00e+00	665
11	NC_069368.1 : 0-end	Chr13 : 0-end	0.00e+00	644
12	NC_069375.1 : 0-end	Chr11 : 0-end	0.00e+00	530
13	NC_069371.1 : 0-end	Chr16 : 0-end	0.00e+00	521
14	NC_069374.1 : 0-end	Chr12 : 0-end	0.00e+00	517
15	NC_069367.1 : 0-end	Chr15 : 0-end	0.00e+00	506
16	NC_069376.1 : 0-end	Chr17 : 0-end	0.00e+00	443
17	NC_069378.1 : 0-end	Chr19 : 0-end	0.00e+00	300
18	NC_069379.1 : 0-end	Chr18 : 0-end	0.00e+00	210
19	NC_048487.1 : 0-end	ctg177 : 0-end	5.01e-07	3
20	NC_048487.1 : 0-end	ctg116 : 0-end	1.46e-03	2

(a) depicts the chromosomes/scaffolds of Mercenaria (x-axis) plotted against the chromosomes/scaffolds of Solidissima (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 14533 orthologs found between 271 Mercenaria scaffolds and 27 Solidissima scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.